

Assignment: Binary Search I

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Easy (6 Questions)

1. Binary Search

Find the index of the target element in a sorted array.

- LeetCode: <https://leetcode.com/problems/binary-search/>
 - GFG: <https://www.geeksforgeeks.org/problems/binary-search-1587115620/1>
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2. Search Insert Position

Find the position where an element should be inserted.

- LeetCode: <https://leetcode.com/problems/search-insert-position/>
 - GFG: <https://www.geeksforgeeks.org/problems/search-insert-position-of-k-in-a-sorted-array/1>
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3. Floor in a Sorted Array

Find the greatest element smaller than or equal to X.

- GFG: <https://www.geeksforgeeks.org/problems/floor-in-a-sorted-array-1587115620/1>
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4. Ceil in a Sorted Array

Find the smallest element greater than or equal to X.

- GFG: <https://www.geeksforgeeks.org/problems/ceil-the-floor2802/1>

5. First Occurrence of X

Return the first index of the target.

- GFG: <https://www.geeksforgeeks.org/problems/first-and-last-occurrences-of-x3116/1>
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6. Last Occurrence of X

Return the last index of the target.

- GFG: <https://www.geeksforgeeks.org/problems/first-and-last-occurrences-of-x3116/1>
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Medium (7 Questions)

1. Find First and Last Position of Element in Sorted Array

Classic lower bound + upper bound application.

- LeetCode: <https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/>
 - GFG: <https://www.geeksforgeeks.org/problems/first-and-last-occurrences-of-x3116/1>
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2. Find Smallest Letter Greater Than Target

Find the next greater character.

- LeetCode: <https://leetcode.com/problems/find-smallest-letter-greater-than-target/>
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3. Count Occurrences in Sorted Array

Use first and last occurrence.

- GFG: <https://www.geeksforgeeks.org/problems/number-of-occurrence2259/1>

4. Peak Index in a Mountain Array

Binary search on sorted/mountain structure.

- LeetCode: <https://leetcode.com/problems/peak-index-in-a-mountain-array/>
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5. Valid Perfect Square

Determine whether a number is a perfect square.

- LeetCode: <https://leetcode.com/problems/valid-perfect-square/>
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6. Guess Number Higher or Lower

Practice binary search template.

- LeetCode: <https://leetcode.com/problems/guess-number-higher-or-lower/>
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7. Search a 2D Matrix

Treat matrix as a sorted array.

- LeetCode: <https://leetcode.com/problems/search-a-2d-matrix/>
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Hard (6 Questions)

1. Find K Closest Elements

Uses lower bound as the starting point.

- LeetCode: <https://leetcode.com/problems/find-k-closest-elements/>
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2. Find Right Interval

Lower bound over interval starts.

- LeetCode: <https://leetcode.com/problems/find-right-interval/>
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3. K-th Smallest Pair Distance

Binary search with counting (advanced).

- LeetCode: <https://leetcode.com/problems/find-k-th-smallest-pair-distance/>
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4. Find in Mountain Array

Interactive binary search.

- LeetCode: <https://leetcode.com/problems/find-in-mountain-array/>
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5. H-Index II

Binary search on sorted citations.

- LeetCode: <https://leetcode.com/problems/h-index-ii/>
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6. Single Element in a Sorted Array

Locate the unique element in $O(\log n)$.

- LeetCode: <https://leetcode.com/problems/single-element-in-a-sorted-array/>